

Project ID :

25-26J-444

1. Topic (12 words max)

FreshRoute

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The fruit supply chain in Sri Lanka suffers from severe structural inefficiencies and a lack of modern technological integration, which significantly affect both producers and consumers. One of the most pressing issues is the lack of real-time demand forecasting and harvest planning tools for fruit farmers. Without access to accurate market intelligence or predictive tools, farmers often rely on assumptions or outdated trends to decide when and how much to harvest. This leads to over-harvesting, market saturation, and ultimately, significant post-harvest waste—a trend that mirrors global statistics, where nearly 45% of fruits are wasted post-harvest due to mismatched supply and demand [1][2].

Another critical challenge is the over-dependence on intermediaries. In Sri Lanka [3], small-scale farmers typically sell their produce through layers of middlemen who control access to markets and dictate pricing. This system erodes the bargaining power of farmers and leads to highly asymmetrical profit distribution, where the final consumer price bears little relationship to what the farmer receives [3][4][5]. The absence of a direct, transparent digital trading mechanism keeps farmers economically vulnerable and unable to optimize their earnings.

Post-harvest fruit spoilage in Sri Lanka remains a major issue, with studies indicating that 30–40% of harvested fruits are lost due to transport delays, poor route planning, inadequate handling, and the absence of cold chain infrastructure [3], [7]. These inefficiencies are especially severe in rural regions, particularly in a context where storage is unavailable or intentionally avoided in favor of a store-less distribution model. Poor handling during loading and transit, along with route mismanagement, further amplifies losses. The lack of coordination between farmers and transport providers — often relying on informal logistics and manual scheduling — creates additional bottlenecks.

Addressing these challenges requires a digitally coordinated transport system that enables efficient route planning, real-time fleet tracking, and structured pickup scheduling. Incorporating AI-based route optimization can reduce travel time and spoilage risk, while digital trip logs and handling checklists improve traceability and reduce mechanical damage. Such an approach directly responds to the inefficiencies identified in the fruit supply chain, where poor logistics significantly reduce product quality and farmer income [3], [8].

In addition, there is an absence of robust quality assurance mechanisms within the fruit supply chain. Without standardized inspection or verification protocols, buyers often receive products that do not meet expected quality levels. Disputes frequently arise between buyers and sellers over the condition of delivered goods, with no objective method to assess responsibility or resolve claims [6]. This lack of trust hinders the development of direct supply relationships and discourages scale-up of decentralized trade models.

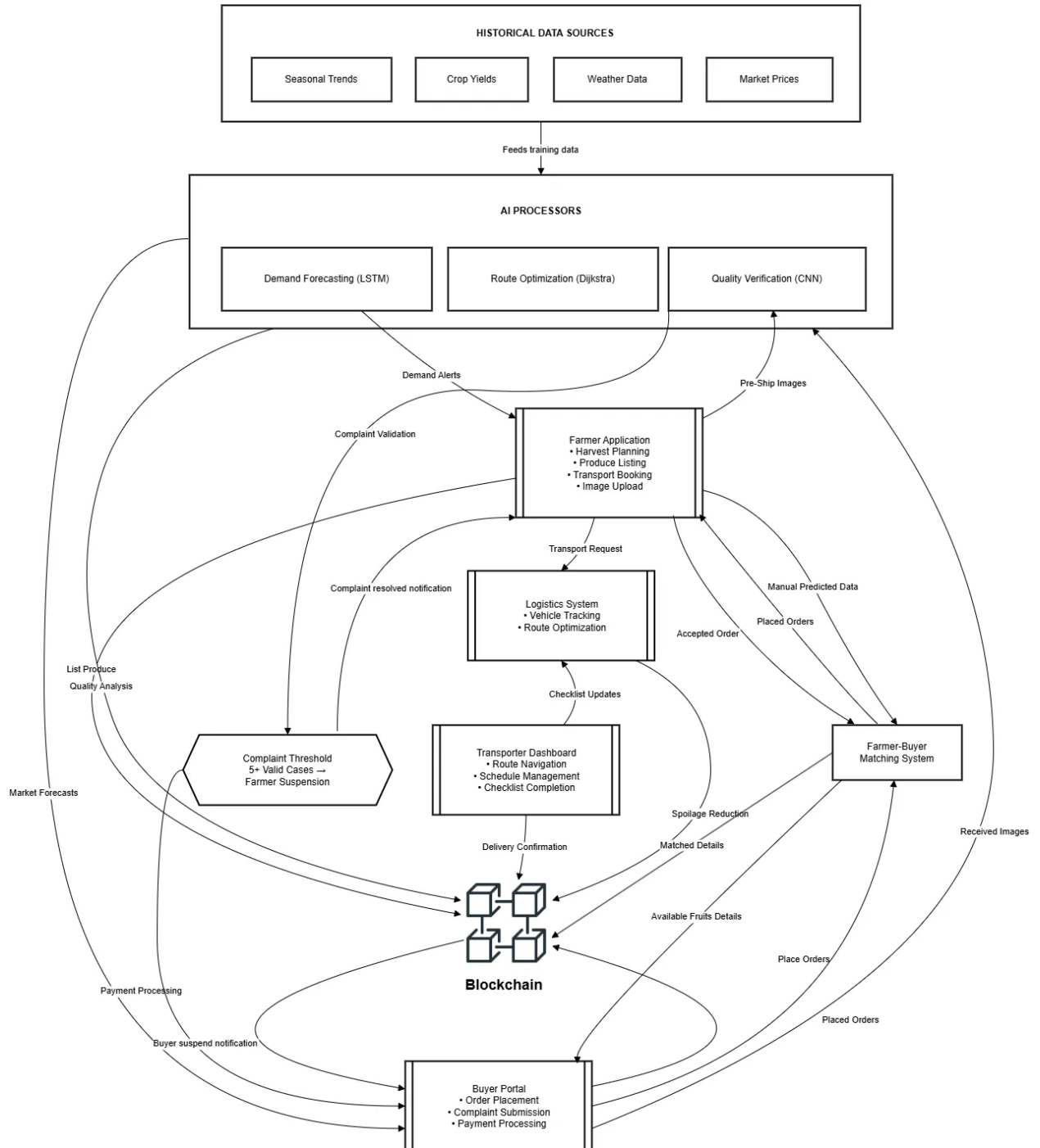
These multifaceted challenges—ranging from forecasting failure, middleman dominance, logistics breakdowns, to quality disputes—highlight a systemic research problem in need of holistic, data-driven transformation to ensure fairness, transparency, and efficiency in Sri Lanka's

- [1] W. Wimalaratana and W. Gammanpila, "Cultivating prosperity in Sri Lanka: Tracing the impact of the fruit supply chain on rural development," *J. Agric. Dev. Stud.*, vol. 38, pp. 1–24, Sep. 19, 2024.
- [2] L. Rajapaksha, D. M. C. C. Gunathilake, S. M. Pathirana, and T. Fernando, "Reducing post-harvest losses in fruits and vegetables for ensuring food security—Case of Sri Lanka," *MOJ Food Process. Technol.*, vol. 9, no. 1, pp. 7–16, 2021.
- [3] C. P. Hathurusinghe, R. Vidanapathirana, R. Rambukwella, and T. G. Somarathne, *A study on value chain of pineapple and banana in Sri Lanka*. Colombo: Hector Kobbekaduwa Agrarian Research and Training Institute, 2012.
- [4] A. K. A. Dissanayake *et al.*, *Export Potential of Major Fruits in Sri Lanka*, Research Report No. 251. Colombo: Hector Kobbekaduwa Agrarian Research and Training Institute, 2022, xv + 136 pp.
- [5] Y. M. P. Samarasinghe, B. A. M. S. Kumara, and A. K. Kulatunga, "Traceability of fruits and vegetables supply chain towards efficient management: A case study from Sri Lanka," *Int. J. Ind. Eng. Oper. Manag.*, vol. 3, no. 02, pp. 89–106, 2021.
- [6] A. L. Sandika, "Impact of middlemen on vegetable marketing channels in Sri Lanka," *Trop. Agric. Res. Ext.*, vol. 14, no. 3, 2012.
- [7] W. M. A. P. Weerasinghe and K. P. K. Premanath, "Quality and Safety Issues in Fruit and Vegetable Supply Chains in Sri Lanka: A Review," *Journal of Food Quality and Hazards Control*, vol. 8, no. 3, pp. 122–128, 2021.
- [8] J. A. Withanage and R. Pathirana, "Safety Issues in Fresh Fruit and Vegetable Supply Chains in Sri Lanka: A Review," *Agricultural Research Journal*, vol. 58, no. 2, pp. 201–209, 2021.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Solution Overview: The Smart Fruit Supply Chain System is a digital platform integrating AI, blockchain, and logistics optimization to transform Sri Lanka's fragmented fruit supply chain. It eliminates structural inefficiencies through four interconnected modules:

1. **AI Demand Forecasting:**
Uses LSTM/Prophet models with historical pricing, weather, and yield data to generate real-time harvest alerts. Farmers receive SMS/push notifications for optimal harvest timing, reducing overproduction and waste.
2. **Blockchain Marketplace:**
Enables direct B2B/B2C trading via smart contracts, bypassing intermediaries. AI matches farmers with buyers based on location, quality needs, and demand. Transparent pricing ensures farmers receive 40-50% higher income.
3. **Spoilage-Reduction Logistics:**
Implements Dijkstra's algorithm for dynamic route optimization. Farmers pre-book transport slots via mobile apps; transporters receive GPS-optimized routes and digital checklists, cutting spoilage by 30-40% through reduced transit time and cold-chain monitoring.
4. **Quality Assurance:**
CNN models compare farmer/buyer-uploaded fruit images to detect damage/spoilage. Validated complaints trigger automatic farmer suspension after 5+ incidents, enforcing accountability.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The successful implementation of the Smart Fruit Supply Chain System requires interdisciplinary expertise across several key domains, including Artificial Intelligence (AI), agriculture and agribusiness, logistics optimization, blockchain technology, and mobile application development.

From an AI perspective, the team must possess a strong understanding of time-series forecasting models such as LSTM (Long Short-Term Memory) and Facebook Prophet, to accurately predict market demand based on historical pricing, seasonal trends, and weather data. Additionally, knowledge in computer vision and convolutional neural networks (CNNs) is required to implement image-based fruit quality verification and automated complaint resolution.

Domain expertise in agricultural economics and supply chain practices is essential to understand the seasonal harvesting patterns, typical post-harvest losses, and how pricing dynamics operate within rural farmer-buyer ecosystems. This includes understanding perishability, buyer behavior, and farmer constraints such as lack of cold storage.

Logistics knowledge is also crucial to develop effective AI-based route optimization models. Familiarity with Dijkstra's algorithm, real-time GPS data integration, and last-mile delivery challenges will support the development of a fleet coordination system to reduce spoilage during transport.

For secure and transparent transactions, blockchain technology is utilized to record farmer-buyer deals. Knowledge in smart contracts, transaction hashing, and decentralized ledgers is necessary.

On the development side, expertise in React Native (or Flutter) for offline-first mobile app creation, Firebase/MongoDB for data syncing, and Node.js/Python for backend logic and AI integration is needed.

Data Requirements:

- Historical fruit price and demand data (by region and type)
- Weather and crop yield records
- GPS and transport route data
- Farmer and buyer profiles
- Image datasets of fresh vs. spoiled fruits for training AI models

This multi-domain integration ensures that the platform is robust, scalable, and directly addresses real-world problems in Sri Lanka's agricultural economy.

8. Objectives and Novelty

Main Objective Creating a software solution process of enabling fruit growers to market their fruit harvest at a fair price without Intermediaries connection and consumers to obtain high quality fresh fruits at reasonable prices using pre-harvest planning and post-harvest damage prevention methods.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Samarakoon A D – IT22124494	Develop a real-time demand forecasting module for harvest planning and implement a demand alerts system	- Collect historical market price and demand data - Collect weather and seasonal yield patterns - Design and train an AI forecasting model (LSTM or Prophet) - Integrate forecast results into a harvest planning dashboard - Validate forecasts with actual market data - Sends real-time alerts for farmers and buyers regarding market demand and warnings.	- First-time use of AI (LSTM/Prophet) for fruit demand forecasting tailored to Sri Lankan market - Offline-capable mobile dashboard for rural farmers - Combines weather, seasonality, and market data for accurate planning - A real-time alert system about fruit demand and harvesting warnings.

Dias N A – IT22213730	Build a direct-to-buyer digital trading platform to eliminate middlemen	- Study current fruit trading patterns and price differences caused by intermediaries - Identifies buying patterns, quality preferences, delivery requirements, and seasonal trends - Design and develop a B2B/B2C digital marketplace for farmers and buyers - Implement AI-based buyer-farmer matching location and product needs - Use blockchain-based secure transaction system	- Blockchain-Enabled Zero-Intermediary Trust Network - Comprehensive Farmer-Buyer Matching - Dual B2B-B2C Integrated Marketplace
Gallearachchi S C - IT22230874	Reduce post-harvest fruit spoilage and transport-related losses by optimizing rural logistics through real-time coordination, AI-based routing, and digital fleet monitoring in Sri Lanka's fruit supply chain.	- Implement AI-powered route optimization using distance, time, and traffic data - Suggest the shortest & fastest pickup routes for collection vehicles - Build a pickup scheduling system for farmers to pre-book collection slots - Enable real-time notifications for transporters via mobile/SMS - Design a fleet management dashboard to track vehicle location and status - Add pre-trip loading checklists and in-app handling tips for transporters - Create a mock interface for cold chain tracking and temperature alerts - Generate digital transport logs with pickup, route, and delivery data	- AI-driven route planning for reducing transport spoilage in Sri Lanka's fruit logistics system - Digitally synchronizes farmers and transporters - Introduces centralized fleet visibility and scheduling, replacing manual systems - Tackles physical fruit damage with digital loading protocols, enhancing post-harvest quality

Dayarathna W S R-IT22257154	Ensure post-harvest fruit quality transparency and trust between farmers and buyers using image-based verification and AI-supported decision-making tool.	- Allow farmers to upload fruit images at the time of packaging before transport - Allow buyers to upload received fruit images upon delivery - Implement an AI-based image comparison tool to detect damage, ripeness level, or spoilage - Create a complaint submission feature for buyers - Automatically validate complaints using AI model output - Maintain complaint history per farmer - If a farmer receives more than 5 valid complaints, automatically suspend the farmer's listings in the system for review	- First use of AI image verification in Sri Lankan fruit supply chain for quality assurance - Buyers and farmers both contribute visual proof, creating mutual accountability - Fully automated reputation-based farmer suspension system based on image evidence and verified complaints
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9. Individual component description of how it is complied with the specialization.

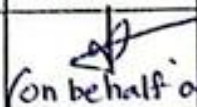
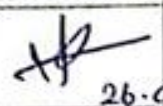
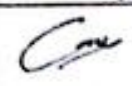
Member Name with Registration No	Description
Samarakoon A D – IT22124494	The component involves building an AI-powered real-time demand forecasting and alert system to support farmers in planning harvests based on actual market needs. This task aligns directly with software engineering specialization through the following: Dual-Interface B2B/B2C Architecture: Applies time-series forecasting models (LSTM/Prophet) using Python libraries like TensorFlow or PyTorch to predict demand based on market price trends, weather, and seasonal yield patterns. Data Engineering: Collects and processes data from external APIs (e.g., weather, market price databases) through custom pipelines, ensuring clean and structured inputs for model training. Mobile Dashboard Development: Builds an offline-first mobile dashboard, allowing rural farmers to access forecasted demand data even in low-connectivity regions. UI/UX Design: Designs a highly intuitive and user-friendly interface tailored for low-tech users. Focuses on minimal interaction flows, large touch targets, readable fonts, icon-based navigation, and offline status indicators. Ensures farmers can easily view demand charts, weather impacts, and alerts without needing training. Alert System Functionality: Integrates a real-time SMS and push notification system (using Firebase or Twilio) to deliver market demand alerts and planning recommendations to farmers and buyers. System Architecture & Integration: Implements a modular backend to handle AI predictions, user data, and alert triggers efficiently. Supports integration with other components like the trading platform and logistics system for future scalability. This component demonstrates the application of AI, mobile development, data

	systems, and human-centered UI/UX design—all core areas of software engineering—toward solving a real-world agricultural problem in Sri Lanka.
Dias N A – IT22213730	This component builds a direct-to-buyer digital trading platform to eliminate intermediaries from the fruit supply chain. It strongly aligns with software engineering principles as follows: Dual-Interface B2B/B2C Architecture & Full-Stack Development: Develops a scalable, multi-tenant platform with dedicated interfaces for farmers and buyers supporting both wholesale and retail transactions. Uses modern full-stack web technologies to create responsive dashboards with unified backend infrastructure. Blockchain-Secured Transaction Framework & Smart Contract Integration: Uses smart contracts (via Ethereum or Hyperledger) to secure transactions, reduce fraud, and maintain tamper-proof trade histories. Automates payment release upon delivery confirmation through cryptographically secured blockchain transactions. Integrated Payment Gateway & Financial Infrastructure: Secure transaction processing through API integration with local payment systems enabling instant payouts to farmers. Features automated escrow services and real-time financial settlements for immediate payment upon delivery verification. UI/UX Design & Multi-Platform Architecture: Designs dual-interface dashboards for farmers and buyers with tailored functionalities. Emphasize simplicity, responsive design, and language localization to support diverse user base. Features include trade notifications, pricing tools, and user reviews. Backend Logic & System Security: Develops user roles (farmer, buyer, admin), authentication, and access control mechanisms. Uses REST APIs and encrypted communication protocols to ensure data integrity and security across all transactions. This component enables traceable, fair trade and empowers farmers with digital tools using solid SE practices like scalable architecture, full-stack dev, smart contract integration, and user-focused design.

Gallearachchi S C - IT22230874	This component addresses transport-related fruit spoilage through a smart logistics module, leveraging multiple software engineering disciplines: AI-Based Route Optimization: Implements dynamic route planning using algorithms via tools like Google OR-Tools and Leaflet for GIS mapping. Scheduling & Notification System: Builds a time-slot reservation system for transport pickup using asynchronous backend logic and push/SMS notifications with event-based architecture. Fleet Management Dashboard: Creates a centralized dashboard to monitor vehicle statuses, locations, and load capacity. Backend APIs sync real-time GPS and sensor data. Data Logging & Traceability: Auto-generates transport logs (timestamps, routes, delivery proofs), enabling audit trails and post-harvest traceability through secure database schemas. Offline Support & Syncing: Uses local caching and queued uploads to handle areas with poor connectivity, syncing updates once reconnected. This module reflects core SE expertise in algorithm design, real-time systems, mobile-responsive UIs, backend APIs, and user-centric fleet management tools for minimizing transport-related fruit losses.
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Dayarathna W S R - IT22257154	This component introduces an AI-integrated image comparison and quality validation system to support a store-less fruit trading platform. It ensures transparency and accountability by validating fruit conditions through images before and after delivery. • AI-Based Image Processing Utilizes CNNs (e.g., MobileNet, EfficientNet) to detect spoilage, ripeness, or damage from images uploaded by farmers and buyers. Enables automated quality verification through AI. • Backend Complaint Management System Develops RESTful APIs to manage complaint workflows, including status tracking, automated escalation, and suspension triggers. • Secure Two-Sided Visual Verification Implements dual image upload for farmers and buyers via Firebase or Cloudinary, ensuring visual proof is stored securely and tied to transaction IDs and timestamps. • Rule-Based Automation and Suspension Applies business rules to auto-flag and suspend farmers with more than 5 verified complaints. Helps maintain trust and product quality within the system. • Database & Record Management Uses a DB for storing complaint logs, image metadata, validation scores, and reputation data. Designed with structured schema for high-performance queries. • System Integration Synchronizes with the marketplace and forecasting dashboards to influence farmer visibility and trade decisions based on quality assurance data. This component reflects core SE practices in AI, full-stack development, rule-based automation, backend API design, cloud integration, and inclusive UI/UX design, all tailored for Sri Lanka's fruit supply chain.
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10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Karthiga	Rajendran	 (on behalf of karthiga)
Co-Supervisor	Ms.	Poojani	Gunathilake	 26.06.2025
External Supervisor	MR.	Chaminda	Gamage	

Summary of external supervisor's (if any) experience and expertise

Farmer:

- A large-scale pineapple farm owner with more than 10 years' experience in pineapple farming and fruit cultivation.
- Deals with intermediaries after the harvesting process.
- Have a good knowledge about fruit quality, fruit grading and marked demand.

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.